DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

B.Tech Degree Mid-Semester Examination — Semester III
CS301: DATA STRUCTURES & ALGORITHMS
Time: 3 Hours | Max Marks: 100

PART A (Answer all 10 questions — 10 x 2 = 20 Marks)

1. Define Big-O notation and provide the tight upper bound of binary search.

2. Differentiate between an Array and a Singly Linked List in terms of random access.

3. Write the prefix form of the infix expression: (A + B) * (C - D) / E.

4. State the balance factor condition for an AVL Tree node.

5. What is the worst-case time complexity of QuickSort and how can it be avoided?

PART B (Answer all 5 questions — 5 x 16 = 80 Marks)

11. (a) Construct an AVL Tree by inserting the following sequence: 40, 20, 10, 25, 30, 22, 50. Show the LL, RR, LR, or RL rotations at every rebalancing step. *(16 Marks)*

12. (a) Explain Dijkstra's Shortest Path algorithm on a weighted directed graph with pseudocode and time complexity analysis using a priority queue (min-heap). *(16 Marks)*

13. (a) Describe the 0/1 Knapsack problem using Dynamic Programming. Tabulate the values for Capacity W=7 and items with weights [2, 3, 4] and values [3, 4, 5]. *(16 Marks)*

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